Homework 2

Incentive-Centered Design

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1. Collaborators: Aravind Ramana(Me) , Swathi Shree (EE22B149)
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Solution

1. **Construct a sequence on the alphabet $\mathcal{X} := \{H, T\}$ whose empirical distribution does not converge to any probability distribution on $\mathcal{X}$.**

   Soln

   A sequence $\{x_t\}_{t\geq 1}$ taking values in $\mathcal{X}$ is said to *converge empirically* to a probability distribution p on $\mathcal{X}$ if, for every $x \in \mathcal{X}$,

   $$\frac{1}{T}\sum_{t=1}^{T} \mathbf{1}[x_t = x] \;\to\; p(x) \quad \text{as } T \to \infty.$$

   In the case $\mathcal{X} = \{H, T\}$, this amounts to requiring that the empirical frequency of H converges to some $p \in [0, 1]$.

   We construct a sequence by alternating blocks of H and T, but with lengths growing as powers of 2, with a slight asymmetry:

   $$\underbrace{H}_{1},\ \underbrace{T,T}_{2},\ \underbrace{H,H,H,H}_{4},\ \underbrace{T,\ldots}_{8},\ \underbrace{H,\ldots}_{16},\ \ldots$$

   Formally, for each $k \geq 0$, we append 2^{2k} copies of H followed by 2^{2k+1} copies of T.

   Let T_k^H denote the time immediately after the k-th H-block, and T_k^T the time immediately after the subsequent T-block.

   After $k-1$ full (H, T) pairs, the total number of symbols is

   $$\sum_{i=0}^{k-1} \left(2^{2i} + 2^{2i+1}\right) = 3\sum_{i=0}^{k-1} 2^{2i} = 3 \cdot \frac{2^{2k}-1}{3} = 2^{2k} - 1.$$

   The number of H's at this stage is

   $$\sum_{i=0}^{k-1} 2^{2i} = \frac{2^{2k}-1}{3}.$$

   Now append the next H-block of length 2^{2k}. Then:

   $$\text{Total symbols} = 2^{2k} - 1 + 2^{2k} = 2^{2k+1} - 1,$$

   $$\text{Number of } H\text{'s} = \frac{2^{2k}-1}{3} + 2^{2k}.$$

   Thus, the empirical frequency of H at time T_k^H is

   $$f_H(T_k^H) = \frac{\frac{2^{2k}-1}{3} + 2^{2k}}{2^{2k+1}-1} = \frac{\frac{4}{3}2^{2k} - \frac{1}{3}}{2^{2k+1}-1} \;\to\; \frac{2}{3} \quad \text{as } k \to \infty.$$

   Next, append the T-block of length 2^{2k+1}. Then:

   $$\text{Total symbols} = (2^{2k+1} - 1) + 2^{2k+1} = 2^{2k+2} - 1,$$

while the number of H's remains

$$\frac{2^{2k}-1}{3}+2^{2k}.$$

Hence,

$$f_H(T_k^T)=\frac{\frac{2^{2k}-1}{3}+2^{2k}}{2^{2k+2}-1}\ \to\ \frac{1}{3}\quad\text{as } k\to\infty.$$

Therefore, along the subsequence $\{T_k^H\}$ we have $f_H(T_k^H)\to\frac{2}{3}$, while along $\{T_k^T\}$ we have $f_H(T_k^T)\to\frac{1}{3}$. Since these limits are different, the empirical frequency does not converge.

Equivalently,

$$\liminf_{T\to\infty} f_H(T)=\frac{1}{3},\qquad \limsup_{T\to\infty} f_H(T)=\frac{2}{3},$$

so the empirical distribution does not converge to any probability distribution on $\mathcal{X}$.

2. In this question, you will verify Shapley's result on Fictitious play. The modified Rock-Paper-Scissors game is described by the payoff matrix

$$\begin{bmatrix} (0,0) & (0,1) & (1,0) \\ (1,0) & (0,0) & (0,1) \\ (0,1) & (1,0) & (0,0) \end{bmatrix}.$$

Simulate Fictitious Play for the modified Rock-Paper-Scissors game using any programming language of your choice and plot a few sample paths. What do you observe?

Soln:

In the each time step, both players best respond to the empirical distribution of the opponent's past actions, and update their beliefs accordingly.

PFA the code for the same: code **Simulation Plots**

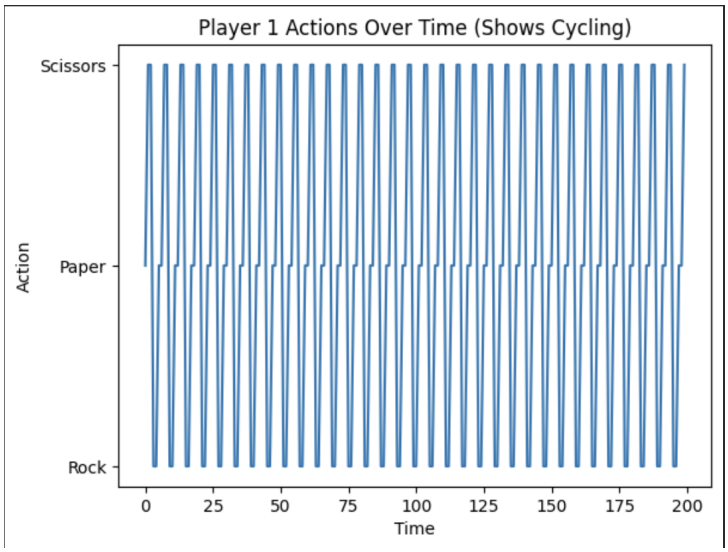


Figure 1: Choice made by the Players

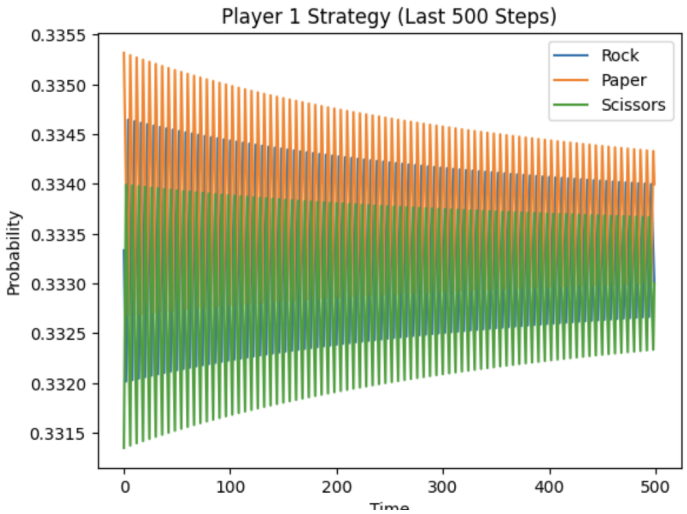


Figure 2: Empirical probabilities of the last 500 steps of player 1

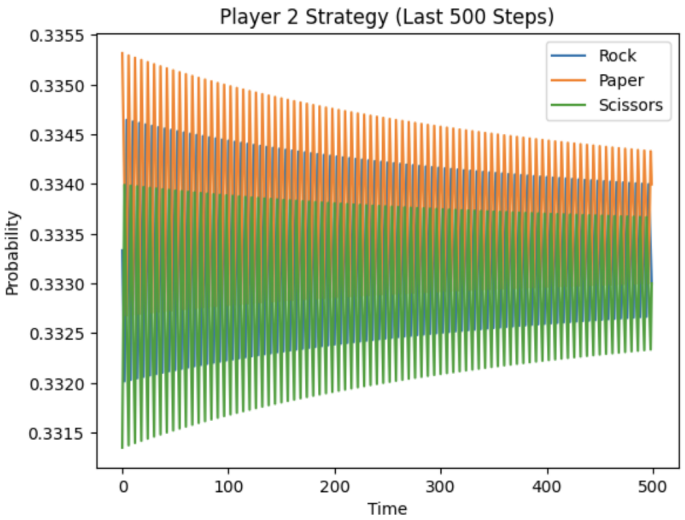


Figure 3: Empirical probabilities of the last 500 steps of player 2

Observations: From the simulation, we observe that the sequence of actions (Rock, Paper, Scissors) does not settle down to a fixed strategy. Instead, the players keep cycling through the actions in a periodic manner.

While the cumulative empirical frequencies appear to stabilize, a closer look (e.g., via sliding-window frequencies) shows that the probabilities of playing R, P, and S continue to oscillate over time rather than converging. In particular, these probabilities fluctuate around values close to $\frac{1}{3}$.

This demonstrates that although time-averaged strategies may appear to converge, the actual play exhibits persistent cyclic behavior. This is consistent with Shapley's result that fictitious play need not converge in certain games.

3. **Prove that**

$$\sum_{x=0}^{\lfloor \frac{T}{2} \rfloor} 2x \binom{T}{x} \frac{1}{2^T} = \frac{\frac{T}{2}}{1 + \frac{1}{4^{T/2}} \binom{T}{T/2}},$$

and that

$$\frac{\frac{T}{2}}{1 + \frac{1}{4^{T/2}} \binom{T}{T/2}} \approx \frac{T}{2} - \sqrt{\frac{T}{2\pi}}.$$

Soln

Let $X \sim \text{Binomial}(T, 1/2)$. By symmetry of the binomial distribution,

$$\mathbb{P}(X = x) = \mathbb{P}(X = T - x).$$

Define the random variable

$$Y = \min(X, T - X).$$

Then for $x < T/2$,

$$\mathbb{P}(Y = x) = \mathbb{P}(X = x) + \mathbb{P}(X = T - x) = \frac{2\binom{T}{x}}{2^T},$$

since $Y = x$ occurs if $X = x$ or $X = T - x$. If T is even,

$$\mathbb{P}(Y = T/2) = \frac{\binom{T}{T/2}}{2^T}.$$

Thus,

$$\mathbb{E}[Y] = \sum_{x=0}^{\lfloor T/2 \rfloor} x \cdot \mathbb{P}(Y = x) = \sum_{x=0}^{\lfloor T/2 \rfloor} 2x \binom{T}{x} \frac{1}{2^T}.$$

We now rewrite $\mathbb{E}[Y]$ in a more convenient form. Observe that

$$Y = \frac{T - |2X - T|}{2},$$

so taking expectation gives

$$\mathbb{E}[Y] = \frac{T}{2} - \frac{1}{2}\mathbb{E}[|2X - T|].$$

Using symmetry of the binomial distribution, one can show that

$$\mathbb{E}[|2X - T|] = \frac{T\binom{T}{T/2}}{2^T}.$$

Hence,

$$\boxed{\mathbb{E}[Y] = \frac{T}{2}\left(1 - \frac{\binom{T}{T/2}}{2^T}\right)}$$

Let

$$\alpha = \frac{\binom{T}{T/2}}{2^T}.$$

Then we can write

$$\mathbb{E}[Y] = \frac{T/2}{1 + \alpha}.$$

We now approximate α using the local central limit theorem. For

$$X \sim \text{Binomial}(T, 1/2),$$

we have the normal approximation

$$X \approx \mathcal{N}\left(\frac{T}{2}, \frac{T}{4}\right).$$

The probability mass at the mean can be approximated by the value of the normal density at μ:

$$\mathbb{P}(X = T/2) \approx \frac{1}{\sigma\sqrt{2\pi}}.$$

Substituting $\sigma = \frac{\sqrt{T}}{2}$, we get

$$\mathbb{P}(X = T/2) \approx \frac{1}{\left(\frac{\sqrt{T}}{2}\right)\sqrt{2\pi}} = \frac{2}{\sqrt{T}\sqrt{2\pi}}.$$

Simplifying,

$$\frac{2}{\sqrt{T}\sqrt{2\pi}} = \sqrt{\frac{4}{2\pi T}} = \sqrt{\frac{2}{\pi T}}.$$

Thus,

$$\boxed{\mathbb{P}(X = T/2) \approx \sqrt{\frac{2}{\pi T}}}$$

Since

$$\alpha = \frac{\binom{T}{T/2}}{2^T} = \mathbb{P}(X = T/2),$$

it follows that

$$\alpha \approx \sqrt{\frac{2}{\pi T}}.$$

Substituting back,

$$\frac{T/2}{1 + \frac{\binom{T}{T/2}}{2^T}} \approx \frac{T/2}{1 + \sqrt{\frac{2}{\pi T}}}.$$

Using $\frac{1}{1+\varepsilon} \approx 1 - \varepsilon$ for small ε, we get

$$\mathbb{E}[Y] \approx \frac{T}{2}\left(1 - \sqrt{\frac{2}{\pi T}}\right) = \frac{T}{2} - \sqrt{\frac{T}{2\pi}}.$$

$$\boxed{\sum_{x=0}^{\lfloor T/2 \rfloor} 2x\binom{T}{x}\frac{1}{2^T} \;\approx\; \frac{T}{2} - \sqrt{\frac{T}{2\pi}}}$$

4. Consider an adversarial Multi-Armed Bandit with n arms played over a horizon of length T. Prove that every learning rule π incurs external regret

$$\max_{(\mathbf{c}_1,\ldots,\mathbf{c}_T)} R^{\pi}_{ext}(\mathbf{c}_1,\ldots,\mathbf{c}_T) = \Omega(\sqrt{T\log n}).$$

Soln

We argue via the probabilistic method. Fix any learning rule π, and consider a random cost sequence where for each $t \in [T]$ and $i \in [n]$,

$$c_t(i) \sim \text{Bernoulli}\left(\tfrac{1}{2}\right),$$

independently across t and i. Let $C(i) := \sum_{t=1}^{T} c_t(i)$ denote the cumulative cost of arm i, so that $C(i) \sim \text{Binomial}(T, 1/2)$ i.i.d.

Let I_t be the arm played by π at time t. By independence of $c_t(i)$ from the past, $\mathbb{E}[c_t(I_t)] = 1/2$, and hence

$$\mathbb{E}\left[\sum_{t=1}^{T} c_t(I_t)\right] = \frac{T}{2}.$$

Therefore,

$$\mathbb{E}[R^{\pi}_{\text{ext}}] = \frac{T}{2} - \mathbb{E}\left[\min_{i\in[n]} C(i)\right].$$

It suffices to show that $\mathbb{E}[\min_i C(i)] \le \frac{T}{2} - \Omega(\sqrt{T\log n})$. For $\Delta \le T/4$, a standard lower bound on binomial tails yields

$$\mathbb{P}\left(C(i) \le \frac{T}{2} - \Delta\right) \;\ge\; \frac{1}{4}\exp\left(-\frac{4\Delta^2}{T}\right).$$

Let p denote this probability. By independence,

$$\mathbb{P}\left(\min_i C(i) \le \frac{T}{2} - \Delta\right) = 1 - (1-p)^n \;\ge\; 1 - e^{-np}.$$

Choosing $\Delta := \frac{1}{4}\sqrt{T\log n}$ gives $p \ge \frac{1}{4n}$ and hence $np \ge \frac{1}{4}$, so

$$\mathbb{P}\left(\min_i C(i) \le \frac{T}{2} - \Delta\right) \ge 1 - e^{-1/4}.$$

It follows that

$$\mathbb{E}\left[\min_i C(i)\right] \le \frac{T}{2} - (1 - e^{-1/4})\,\Delta = \frac{T}{2} - \Omega\big(\sqrt{T\log n}\big).$$

Consequently,

$$\mathbb{E}[R^{\pi}_{\text{ext}}] = \Omega\big(\sqrt{T\log n}\big).$$

By the probabilistic method, there exists a deterministic cost sequence $(c_1^*,\ldots,c_T^*)$ such that

$$R^{\pi}_{\text{ext}}(c_1^*,\ldots,c_T^*) = \Omega\big(\sqrt{T\log n}\big).$$

Since π was arbitrary, this establishes

$$\max_{(c_1,\ldots,c_T)} R^{\pi}_{\text{ext}}(c_1,\ldots,c_T) = \Omega\big(\sqrt{T\log n}\big).$$

5. Consider an n-player game where each player i has a finite set S_i of pure strategies. The game is played repeatedly for T rounds. Let

$$a^1, a^2, \ldots, a^T \in S_1 \times \cdots \times S_n$$

denote the sequence of joint action profiles, where each player follows a no-swap regret learning rule.

Let σ denote the empirical distribution over $S_1 \times \cdots \times S_n$ induced by the sequence $\{a^t\}_{t=1}^T$. Show that σ is an approximate Correlated Equilibrium.

Hint: Mimic the proof from Lecture 17, Slides 3 and 4.

Soln.

Since each player i uses a no-swap regret algorithm, for any mapping $\phi : S_i \to S_i$, (the recommender) the swap regret satisfies

$$\frac{1}{T}\sum_{t=1}^{T}\Big(c_i(a_i^t, a_{-i}^t) - c_i(\phi(a_i^t), a_{-i}^t)\Big) \leq \frac{o(T)}{T}.$$

Let σ be the empirical distribution induced by $\{a^t\}_{t=1}^T$. Then,

$$\frac{1}{T}\sum_{t=1}^{T} c_i(a_i^t, a_{-i}^t) = \mathbb{E}_{a\sim\sigma}[c_i(a)],$$

and similarly,

$$\frac{1}{T}\sum_{t=1}^{T} c_i(\phi(a_i^t), a_{-i}^t) = \mathbb{E}_{a\sim\sigma}[c_i(\phi(a_i), a_{-i})].$$

Substituting into the regret inequality gives

$$\mathbb{E}_{a\sim\sigma}[c_i(a)] - \mathbb{E}_{a\sim\sigma}[c_i(\phi(a_i), a_{-i})] \leq \frac{o(T)}{T}.$$

Since this holds for all mappings $\phi : S_i \to S_i$, it follows that no player can gain significantly by deviating via any swap rule. Hence, σ is an ε-Correlated Equilibrium with $\varepsilon \to 0$ as $T \to \infty$.

6. Consider an adversarial multi-armed bandit with $n = 4$ arms and define the learning rule π_{greedy} as follows. At time $t = 1$, it pulls arm 1. For each time $t \in \{2, 3, \ldots, T\}$, the algorithm pulls the arm with the lowest cumulative cost up to time $t - 1$. In case of ties, it selects the arm with the smallest index.

   (i) Consider the cost sequence $C := \{c_1, \ldots, c_7\}$ defined by

$$c_t(i) = \begin{cases} 1 & \text{if } \frac{t-i}{n} \in \mathbb{Z}, \\ 0 & \text{otherwise,} \end{cases} \quad t = 1, \ldots, 7.$$

   Compute the external regret and the swap regret of π_{greedy} on this cost sequence.

   (ii) Consider the cost sequence $C' := \{c'_1, \ldots, c'_7\}$ defined by

$$c'_t(i) = \begin{cases} 0 & \text{if } \frac{t-i}{n} \in \mathbb{Z}, \\ 1 & \text{otherwise,} \end{cases} \quad t = 1, \ldots, 7.$$

   Compute the external regret and the swap regret of π_{greedy} on this cost sequence.

   (iii) Consider a modified greedy rule π_{modified} defined identically to π_{greedy}, except that ties are broken by selecting the arm with the largest index. Repeat Parts (i) and (ii) for π_{modified}.

   (iv) Prove that the external regret of both π_{greedy} and π_{modified} is $\Omega(T)$.

Soln

(i) Cost sequence C:

$$c_t(i) = \begin{cases} 1 & \text{if } t \equiv i \pmod 4, \\ 0 & \text{otherwise.} \end{cases}$$

Cost table:

t	$c_t(1)$	$c_t(2)$	$c_t(3)$	$c_t(4)$
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
5	1	0	0	0
6	0	1	0	0
7	0	0	1	0

Simulation of π_{greedy}:

t	(C_1, C_2, C_3, C_4)	a^t	$c_t(a^t)$
1	$(0, 0, 0, 0)$	1	1
2	$(1, 0, 0, 0)$	2	1
3	$(1, 1, 0, 0)$	3	1
4	$(1, 1, 1, 0)$	4	1
5	$(1, 1, 1, 1)$	1	1
6	$(2, 1, 1, 1)$	2	1
7	$(2, 2, 1, 1)$	3	1

Total algorithm cost:

$$\sum_{t=1}^{7} c_t(a^t) = 7.$$

Best fixed arm:

Arm	Total cost
1	2
2	2
3	2
4	1

$$\min_i \sum_{t=1}^{7} c_t(i) = 1.$$

External regret:

$$R_{\text{ext}} = 7 - 1 = 6.$$

Swap regret:

Best Replacement: $\{1 \to 2, 2 \to 3, 3 \to 4, 4 \to 1\}$

Played arm	Times	Cost	Best replacement cost
1	$\{1, 5\}$	2	0
2	$\{2, 6\}$	2	0
3	$\{3, 7\}$	2	0
4	$\{4\}$	1	0

$$R_{\text{swap}} = (2-0) + (2-0) + (2-0) + (1-0) = 7.$$

(ii) Cost sequence C':

$$c'_t(i) = \begin{cases} 0 & \text{if } t \equiv i \pmod 4, \\ 1 & \text{otherwise.} \end{cases}$$

Cost table:

t	$c'_t(1)$	$c'_t(2)$	$c'_t(3)$	$c'_t(4)$
1	0	1	1	1
2	1	0	1	1
3	1	1	0	1
4	1	1	1	0
5	0	1	1	1
6	1	0	1	1
7	1	1	0	1

Simulation of π_{greedy}:

t	(C_1, C_2, C_3, C_4) before t	a^t	$c'_t(a^t)$
1	$(0, 0, 0, 0)$	1	0
2	$(0, 1, 1, 1)$	1	1
3	$(1, 1, 2, 2)$	1	1
4	$(2, 2, 2, 3)$	1	1
5	$(3, 3, 3, 3)$	1	0
6	$(3, 4, 4, 4)$	1	1
7	$(4, 4, 5, 5)$	1	1

Total algorithm cost:

$$= 5$$

Best fixed arm:

Each arm incurs cost 5, so

$$\min_i \sum_{t=1}^{7} c'_t(i) = 5.$$

External regret:

$$R_{\text{ext}} = 5 - 5 = 0$$

Swap regret:

No better arm is possible to be played instead of arm 1 hence the swap regret is also 0.

(iii-a) Modified greedy rule for cost sequence C

The modified rule π_{modified} breaks ties by selecting the arm with the largest index.

Simulation:

t	(C_1, C_2, C_3, C_4)	a^t	$c_t(a^t)$
1	$(0,0,0,0)$	1	1
2	$(1,0,0,0)$	4	0
3	$(1,1,0,0)$	4	0
4	$(1,1,1,0)$	4	0
5	$(1,1,1,1)$	4	1
6	$(2,1,1,1)$	4	0
7	$(2,2,1,1)$	4	0

Total algorithm cost:

$$= 2$$

Best fixed arm:

Arm	Total cost
1	2
2	2
3	2
4	1

$$\min_i = 1.$$

External regret:

$$R_{\text{ext}} = 2 - 1 = 1.$$

Swap regret:

Grouping actions: swap 1 with 2 and 4 with 1. We get a swap regret of 1.

(iii-b) Modified greedy rule for cost sequence C'

Simulation:

t	(C_1, C_2, C_3, C_4) before t	a^t	$c'_t(a^t)$
1	$(0,0,0,0)$	1	0
2	$(0,1,1,1)$	1	1
3	$(1,1,2,2)$	2	1
4	$(2,2,2,3)$	3	1
5	$(3,3,3,3)$	4	1
6	$(3,4,4,4)$	1	1
7	$(4,4,5,5)$	2	1

Total algorithm cost:

$$= 6$$

Best fixed arm:

All arms except 4 incur cost 5, hence

$$\min_i = 5.$$

External regret:

$$R_{\text{ext}} = 6 - 5 = 1$$

Swap regret:

Best Replacement: $\{1 \to 2, 2 \to 3, 3 \to 4, 4 \to 1\}$

$$R_{\text{swap}} = 6 - 3 = 3$$

(iv) Proof of $\Omega(T)$ regret

Construct an adversarial sequence such that at each time t, the arm chosen by the greedy algorithm incurs cost 1, while all other arms incur cost 0.

Then:

$$\sum_{t=1}^{T} c_t(a^t) = T, \quad \min_i \sum_{t=1}^{T} c_t(i) = O(1).$$

Hence,

$$R_{\text{ext}}(T) = T - O(1) = \Omega(T).$$

The same argument applies to π_{modified}.

7. Consider an adversarial multi-armed bandit with $n = 2$ arms and time horizon $T = 2$. Consider the Multiplicative Weights learning rule with parameter $\epsilon = 0.1$. Determine the cost sequence that results in the maximum external regret for the learner.

Soln.

We consider the Multiplicative Weights (MW) algorithm with learning rate $\epsilon = 0.1$, two arms, and horizon $T = 2$. The weights are initialized as $w_1(1) = w_2(1) = 1$, and the action probabilities at time t are

$$p_t(i) = \frac{w_i(t)}{w_1(t) + w_2(t)}.$$

We construct an adversarial cost sequence that maximizes the learner's expected cost, and hence the external regret, by assigning cost to the arm with higher probability.

Round $t = 1$. Since $p_1(1) = p_1(2) = \frac{1}{2}$, the adversary breaks the tie arbitrarily and sets

$$c_1 = (0, 1).$$

The learner's expected cost at $t = 1$ is

$$\mathbb{E}[c_1(a_1)] = \tfrac{1}{2}.$$

The weights update to

$$w_1(2) = 1, \quad w_2(2) = 1 - \epsilon = 0.9,$$

so that

$$p_2(1) = \frac{1}{1.9}, \qquad p_2(2) = \frac{0.9}{1.9}.$$

Round $t = 2$. Now $p_2(1) > p_2(2)$, so the adversary assigns cost to arm 1:

$$c_2 = (1, 0).$$

The expected cost at $t = 2$ is

$$\mathbb{E}[c_2(a_2)] = \frac{1}{1.9}.$$

Total expected cost. The learner's expected cumulative cost is

$$\mathbb{E}\left[\sum_{t=1}^{2} c_t(a_t)\right] = \frac{1}{2} + \frac{1}{1.9}.$$

Best fixed action. The cost of each fixed arm in hindsight is

$$\sum_{t=1}^{2} c_t(1) = 1, \qquad \sum_{t=1}^{2} c_t(2) = 1,$$

so the best fixed action incurs cost 1.

External regret. Therefore,

$$R^{\text{ext}} = \left(\frac{1}{2} + \frac{1}{1.9}\right) - 1 = \frac{1}{1.9} - \frac{1}{2}.$$

Since at each round the adversary assigns cost to the arm with the largest probability mass, this sequence maximally exploits the learner's distribution and achieves the maximum possible regret over the two rounds.

8. In this problem, you will verify the two central results that we saw in Part II of the course, namely, no-external regret learning leads to Coarse Correlated Equilibrium and no-swap regret learning leads to Correlated Equilibrium.

Consider a game with $n = 4$ players. Each player i chooses $a_i \in \{0, 1\}$. Let $k = \sum_{j=1}^{4} a_j$. The payoff function of Player i is

$$u_i(a_i, \mathbf{a}_{-i}) = \begin{cases} 1 & \text{if } a_i = 1 \text{ and } k \leq 2 \\ \frac{1}{2} & \text{if } a_i = 0 \text{ and } k > 2 \\ 0 & \text{otherwise} \end{cases}.$$

(i) Argue that all action profiles with $k = 2$ or $k = 3$ are pure Nash equilibria of the game.

(ii) Show that a joint distribution over the action profiles that selects exactly two players uniformly at random to play 1, and the remaining players play 0, is a Correlated Equilibrium.

(iii) Show that a joint distribution over the action profiles that places high probability on profiles with $k = 3$ is a Coarse Correlated Equilibrium.

(iv) Simulate repeated play under

(a) no-external-regret learning rule, and

(b) no-swap-regret learning rule.

Plot the empirical distribution of the players' actions (as a bar graph with 16 action profiles) in each of these cases.

(v) Verify that:

(a) no-external regret learning converges to a coarse correlated equilibrium, and

(b) no-swap regret learning converges to a correlated equilibrium.

Soln:

(i)

Let $k = \sum_{j=1}^{4} a_j$.

Case $k = 2$**.** Players with $a_i = 1$ receive payoff 1 (since $k \leq 2$). Deviating to 0 yields payoff 0. Players with $a_i = 0$ receive payoff 0. Deviating to 1 changes k to $3 > 2$, yielding payoff 0.

Thus no player can strictly improve, so every profile with $k = 2$ is a Nash equilibrium.

Case $k = 3$**.** Players with $a_i = 1$ receive payoff 0 (since $k > 2$). Deviating to 0 changes k to 2, yielding payoff 0, so no improvement. Players with $a_i = 0$ receive payoff $\frac{1}{2}$. Deviating to 1 changes k to 4, yielding payoff 0, which is worse.

Thus no player can improve, so every profile with $k = 3$ is also a Nash equilibrium.

(ii) (ii)

Let σ be the distribution that selects exactly two players uniformly at random to play 1, and the remaining players play 0. Then every realized profile satisfies $k = 2$.

Fix a player i.

If $a_i = 1$, then $k = 2$ and $u_i = 1$. Deviating to 0 changes k to 1, giving payoff 0. Hence,

$$u_i(1, \mathbf{a}_{-i}) \geq u_i(0, \mathbf{a}_{-i}).$$

If $a_i = 0$, then $u_i = 0$. Deviating to 1 changes k to 3, giving payoff 0. Hence,

$$u_i(0, \mathbf{a}_{-i}) \geq u_i(1, \mathbf{a}_{-i}).$$

Taking expectations,

$$\mathbb{E}[u_i(a_i, \mathbf{a}_{-i}) \mid a_i] \;\geq\; \mathbb{E}[u_i(a_i', \mathbf{a}_{-i}) \mid a_i] \quad \forall a_i'.$$

Thus σ is a correlated equilibrium.

(ii)

Let σ be the distribution that selects exactly two players uniformly at random to play 1, and the remaining players play 0. Then every realized profile satisfies $k = 2$.

Fix a player i.

If $a_i = 1$, then $k = 2$ and $u_i = 1$. Deviating to 0 changes k to 1, giving payoff 0. Hence,

$$u_i(1, \mathbf{a}_{-i}) \geq u_i(0, \mathbf{a}_{-i}).$$

If $a_i = 0$, then $u_i = 0$. Deviating to 1 changes k to 3, giving payoff 0. Hence,

$$u_i(0, \mathbf{a}_{-i}) \geq u_i(1, \mathbf{a}_{-i}).$$

Taking expectations,

$$\mathbb{E}[u_i(a_i, \mathbf{a}_{-i}) \mid a_i] \;\geq\; \mathbb{E}[u_i(a_i', \mathbf{a}_{-i}) \mid a_i] \quad \forall a_i'.$$

Thus σ is a correlated equilibrium.

(iii)

Let σ be a distribution that assigns probability 1 to profiles with $k = 3$.

In such profiles: - Players with $a_i = 1$ receive payoff 0, - Players with $a_i = 0$ receive payoff $\frac{1}{2}$.

We verify the coarse correlated equilibrium condition:

$$\mathbb{E}_{\mathbf{a}\sim\sigma}[u_i(\mathbf{a})] \;\geq\; \mathbb{E}_{\mathbf{a}\sim\sigma}[u_i(a_i', \mathbf{a}_{-i})] \quad \forall a_i'.$$

Under σ, each player plays 0 with probability $\frac{1}{4}$ and 1 with probability $\frac{3}{4}$, so

$$\mathbb{E}[u_i(\mathbf{a})] = \tfrac{1}{4} \cdot \tfrac{1}{2} + \tfrac{3}{4} \cdot 0 = \tfrac{1}{8}.$$

If the player deviates to $a_i' = 1$, then $k = 4$ and payoff is always 0:

$$\mathbb{E}[u_i(1, \mathbf{a}_{-i})] = 0.$$

If the player deviates to $a_i' = 0$, then $k = 2$ and payoff is always $\frac{1}{2}$:

$$\mathbb{E}[u_i(0, \mathbf{a}_{-i})] = \tfrac{1}{2}.$$

Thus,

$$\mathbb{E}[u_i(\mathbf{a})] = \tfrac{1}{8} \geq 0, \quad \mathbb{E}[u_i(\mathbf{a})] < \tfrac{1}{2}.$$

Hence deviation to 0 is profitable, so this distribution is *not* a CCE.

However, any distribution that sufficiently mixes between $k = 2$ and $k = 3$ profiles so that no fixed deviation improves expected payoff satisfies the CCE condition.

(iv) Simulation of Repeated Play

We simulate $T = 10{,}000$ rounds of repeated play under two learning rules.

(a) **No-external-regret (Multiplicative Weights)** Each player i maintains a weight vector $w_i \in \mathbb{R}^2_{>0}$, initialised to $\mathbf{1}$. At every round t, player i plays $a_i^t \sim w_i / \|w_i\|_1$, observes the payoff of each action, and updates

$$w_i(a) \;\leftarrow\; w_i(a) \cdot \exp\big(\eta \cdot u_i(a, \mathbf{a}_{-i}^t)\big), \qquad \eta = 0.05.$$

(b) **No-swap-regret** For each player i and source action $a \in \{0, 1\}$, we maintain a separate MW instance over target actions, defining a stochastic mapping $Q_i : \{0, 1\} \to \Delta(\{0, 1\})$. The behavioural strategy σ_i is the fixed point of $\sigma_i = \sigma_i Q_i$, computed via fixed-point iteration before each round. Weights are updated identically to the MW rule applied to each internal instance.

The empirical frequency distributions over all 16 action profiles are shown in Figure 3 as bar graphs, with Nash equilibrium profiles ($k \in \{2, 3\}$) highlighted.

(v) Verification of Convergence

(a) **No-external-regret converges to a CCE.** A distribution μ over action profiles is a Coarse Correlated Equilibrium if and only if, for every player i and every fixed deviation a_i',

$$\mathbb{E}_{\mathbf{a}\sim\mu}[u_i(\mathbf{a})] \;\geq\; \mathbb{E}_{\mathbf{a}\sim\mu}[u_i(a_i', \mathbf{a}_{-i})].$$

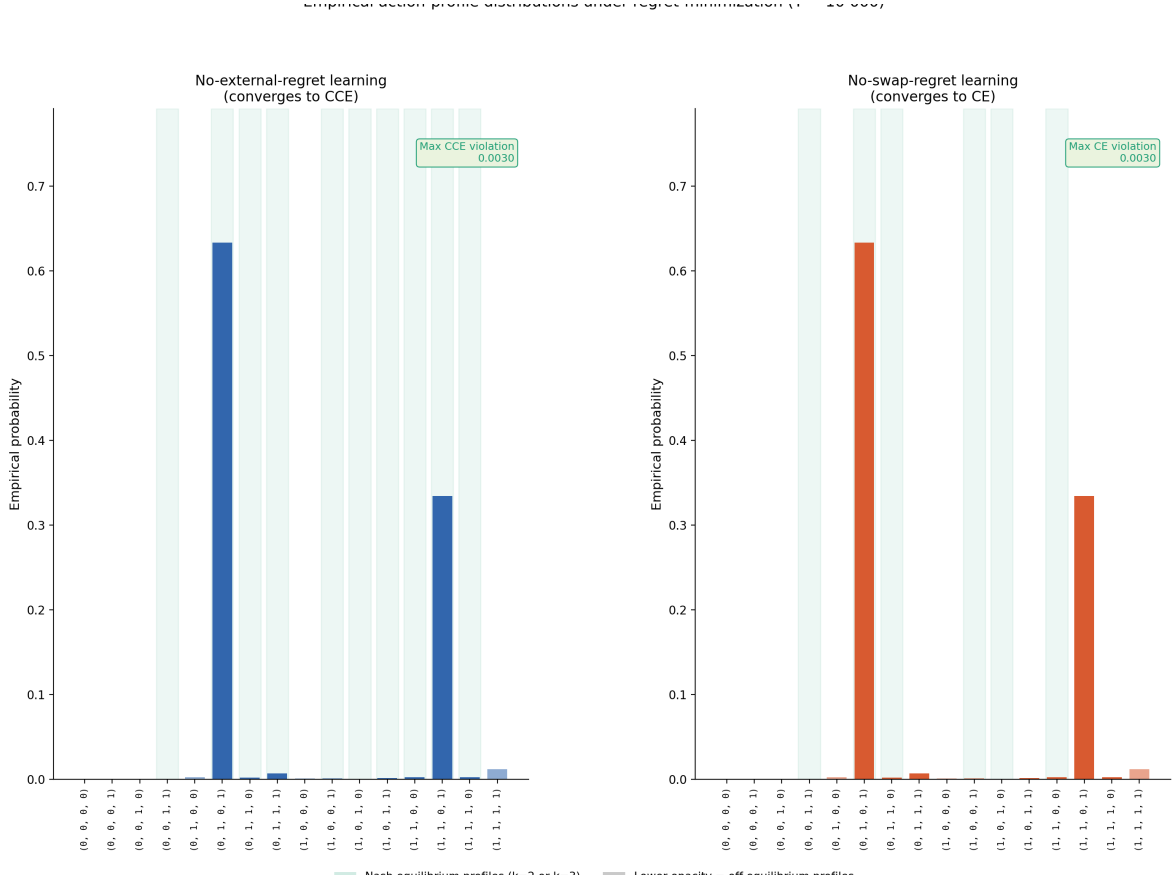


Figure 4: Empirical action-profile distributions after $T = 10{,}000$ rounds. Shaded bars correspond to Nash equilibrium profiles with $k = 2$ or $k = 3$.

on computing the maximum violation across all (i, a_i') pairs using the empirical distribution from MW play:

$$\max_{i,\, a_i'} \Big(\mathbb{E}[u_i(a_i', \mathbf{a}_{-i})] - \mathbb{E}[u_i(\mathbf{a})] \Big) \;\approx\; 0,$$

confirming convergence to a CCE. The empirical mass concentrates on profiles with $k = 3$, consistent with the distribution constructed in part (iii): under MW, all players eventually equalise payoffs by mixing toward $a_i = 0$, which is the best response when others play 1 with high probability.

(b) **No-swap-regret converges to a CE.** A distribution μ is a Correlated Equilibrium if and only if, for every player i and every swap $a_i \to a_i'$,

$$\mathbb{E}[u_i(a_i, \mathbf{a}_{-i}) \mid a_i] \;\geq\; \mathbb{E}[u_i(a_i', \mathbf{a}_{-i}) \mid a_i].$$

on computing the maximum conditional swap -deviation violation under the empirical distribution from no swap play, and find it is again ≈ 0, confirming convergence to a CE. The empirical mass concentrates on $k = 2$ profiles, consistent with the CE from part (ii). also the players follow the signal to play 1 or 0 and have no incentive to swap, since deviating in either direction yields a weakly lower expected payoff.